Data Science

Statistics | Confidence Intervals

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Data Science Track

Confidence Intervals

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Note: Some parts of the content were generated using AI tools and have undergone thorough review and verification by the expert SHAI team to ensure accuracy.

Table of Content

t-distribution

	Confidence Level									
	60%	70%	80%	85%	90%	95%	98%	99%	99.8%	99.9%
	Level of Significance									
2 Tailed	0.40	0.30	0.20	0.15	0.10	0.05	0.02	0.01	0.002	0.001
1 Tailed	0.20	0.15	0.10	0.075	0.05	0.025	0.01	0.005	0.001	0.0005
df										
1	1.376	1.963	3.133	4.195	6.320	12.69	31.81	63.67	—	—
2	1.060	1.385	1.883	2.278	2.912	4.271	6.816	9.520	19.65	26.30
3	0.978	1.250	1.637	1.924	2.352	3.179	4.525	5.797	9.937	12.39
4	0.941	1.190	1.533	1.778	2.132	2.776	3.744	4.596	7.115	8.499
5	0.919	1.156	1.476	1.699	2.015	2.570	3.365	4.030	5.876	6.835
6	0.906	1.134	1.440	1.650	1.943	2.447	3.143	3.707	5.201	5.946
7	0.896	1.119	1.415	1.617	1.895	2.365	2.999	3.500	4.783	5.403
8	0.889	1.108	1.397	1.592	1.860	2.306	2.897	3.356	4.500	5.039
9	0.883	1.100	1.383	1.574	1.833	2.262	2.822	3.250	4.297	4.780
10	0.879	1.093	1.372	1.559	1.813	2.228	2.764	3.170	4.144	4.586
11	0.875	1.088	1.363	1.548	1.796	2.201	2.719	3.106	4.025	4.437
12	0.873	1.083	1.356	1.538	1.782	2.179	2.682	3.055	3.930	4.318
13	0.870	1.079	1.350	1.530	1.771	2.160	2.651	3.013	3.852	4.221
14	0.868	1.076	1.345	1.523	1.761	2.145	2.625	2.977	3.788	4.141
15	0.866	1.074	1.341	1.517	1.753	2.131	2.603	2.947	3.733	4.073
16	0.865	1.071	1.337	1.512	1.746	2.120	2.584	2.921	3.687	4.015
17	0.863	1.069	1.333	1.508	1.740	2.110	2.567	2.899	3.646	3.965
18	0.862	1.067	1.330	1.504	1.734	2.101	2.553	2.879	3.611	3.922
19	0.861	1.066	1.328	1.500	1.729	2.093	2.540	2.861	3.580	3.884
20	0.860	1.064	1.325	1.497	1.725	2.086	2.529	2.846	3.552	3.850
21	0.859	1.063	1.323	1.494	1.721	2.080	2.518	2.832	3.528	3.820
22	0.858	1.061	1.321	1.492	1.717	2.074	2.509	2.819	3.505	3.792
23	0.857	1.060	1.319	1.489	1.714	2.069	2.500	2.808	3.485	3.768
24	0.857	1.059	1.318	1.487	1.711	2.064	2.493	2.797	3.467	3.746
25	0.856	1.058	1.316	1.485	1.708	2.060	2.486	2.788	3.451	3.725
26	0.856	1.058	1.315	1.483	1.706	2.056	2.479	2.779	3.435	3.707
27	0.855	1.057	1.314	1.482	1.703	2.052	2.473	2.771	3.421	3.690
28	0.855	1.056	1.313	1.480	1.701	2.048	2.468	2.764	3.409	3.674
29	0.854	1.055	1.311	1.479	1.699	2.045	2.463	2.757	3.397	3.660
30	0.854	1.055	1.310	1.477	1.697	2.042	2.458	2.750	3.386	3.646
40	0.851	1.050	1.303	1.468	1.684	2.021	2.424	2.705	3.307	3.551
50	0.849	1.047	1.299	1.462	1.676	2.009	2.404	2.678	3.262	3.496
60	0.848	1.045	1.296	1.458	1.671	2.000	2.391	2.661	3.232	3.460
70	0.847	1.044	1.294	1.456	1.667	1.994	2.381	2.648	3.211	3.435
80	0.846	1.043	1.292	1.453	1.664	1.990	2.374	2.639	3.196	3.417
90	0.846	1.042	1.291	1.452	1.662	1.987	2.369	2.632	3.184	3.402
100	0.845	1.042	1.290	1.451	1.660	1.984	2.365	2.626	3.174	3.391
∞	0.842	1.036	1.282	1.440	1.645	1.960	2.327	2.576	3.091	3.291

LEARNING OBJECTIVE

This chapter will explore confidence intervals for population proportions, differences in proportions, population means, and differences in means, highlighting their calculation and interpretation.

By the time you reach the end of this chapter, you will be able to:

1. Explain the concept of population proportion and its significance in estimating characteristics of a population based on sample data.
2. Calculate the confidence interval for a population proportion using the observed sample proportion, standard error, and t-multiplier.
3. Assess the precision of a confidence interval for population proportion by analyzing the margin of error and the confidence level.
4. Describe the role of confidence intervals in estimating a population mean using sample data and explain the factors that affect the margin of error.
5. Calculate the confidence interval for the population mean using the sample mean,

GETTING STARTED

A Researcher's Dilemma

Imagine Lina, a researcher in Amman, tasked with studying the average height of university students.

Measuring every student's height would be impractical, so Lina selects a random sample of 150 students—85 females and 65 males.

standard deviation, and sample size, applying the appropriate t-multiplier.

6. Interpret the confidence interval for the population mean, evaluating how well it represents the true mean based on the given sample data.
7. Define the difference in population proportions and explain how confidence intervals can be used to compare proportions between two groups.
8. Calculate the confidence interval for the difference between two population proportions, using the sample proportions and standard errors of each group.
9. Interpret the confidence interval for the difference in proportions, making conclusions about the relationship between the two groups based on the calculated range.
10. Explain the concept of confidence intervals for the difference in population means and how they are used to compare the average values between two groups.
11. Calculate the confidence interval for the difference in means using the sample means, standard deviations, and sample sizes of both groups, along with the appropriate t-multiplier.
12. Assess the confidence interval for the difference in means and evaluate its implications for comparing two population

After calculating the sample averages, she wonders:

- Can these results represent the entire population of university students in Amman?
- What range of values could reasonably capture the true average height of all students?
- Is there a significant difference in the average heights of male and female students?

To answer these questions, Lina uses confidence intervals, a statistical method that provides a range of plausible values for a population parameter, such as a proportion or a mean, based on sample data.

means, considering the width of the interval and statistical significance.

Confidence Intervals

Starter :

Confidence intervals are a calculated range or boundary around a parameter or a statistic that is supported mathematically with a certain level of confidence. For example, we estimate, with 95% confidence, that the population proportion of teenagers who play video games is somewhere between 32.2% and 37.7%. This is different than having a 95% probability that the true population proportion is within our confidence interval. Essentially, If we were to repeat the sampling process, and calculate a confidence interval for that proportion from each sample , 95% of our calculated confidence intervals would contain the true proportion.

Figure #: " You definitely know who lost the game!"

Confidence interval: a statistical tool used to estimate the range of plausible values for a population parameter (such as a mean or proportion) based on sample data.

Descriptive vs. Inferential Statistics

Descriptive statistics summarizes and organizes data.

- It includes measures of central tendency (mean, median, mode), measures of dispersion (variance, standard deviation), and graphical representations.

Inferential statistics draws conclusions and makes predictions about a population using sample data.

- It involves probability theory, hypothesis testing, confidence intervals, and regression analysis.

How are Confidence Intervals Calculated?

Our equation for calculating confidence intervals is as follows:

$$Best\ Estimate\ \pm\ Margin\ of\ Error$$

Where:

The Best Estimate: is the observed population proportion or mean.

The Margin of Error: is the t-multiplier (t*) times the Standard Error.

The t-multiplier is calculated based on the degrees of freedom and desired confidence level.

For samples with more than 30 observations and a confidence level of 95%, the t-multiplier approximately is 1.96

The equation to create a 95% confidence interval can also be shown as:

$$Population\ Proportion\ or\ Mean\ \pm\ (t^{*}\ \times\ Standard\ Error)$$

The Standard Error is calculated differently for population proportion and mean:

1. Confidence Intervals for Population Proportions

A **population proportion** represents the fraction of a population exhibiting a specific characteristic. Confidence intervals help estimate this proportion within a specified confidence level, such as 95%.

Formula for Standard Error of a Proportion:

$$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Where:

$\hat{p}$: is the sample proportion.

n: is the sample size.

Brain power

Confidence intervals provide a range within which the true population proportion is likely to fall. But what if Lina's sample was collected only from a sports academy or a region known for taller individuals?

👉 How does the way we collect data influence the reliability of our confidence interval?

Example: Proportion of Tall Students


"Not short, just further from the sun!"

Lina observes that 90 of the 150 students in her sample are taller than 170 cm.

The sample proportion is:

$$\hat{p} = \frac{90}{150} = 0.6$$

For a 95% confidence interval, the t-multiplier is 1.96, and the Standard Error is:

$$\sqrt{\frac{0.6\times(1-0.6)}{150}} = 0.04$$

Therefore, the margin of error is:

$$1.96\times0.04 = 0.0784$$

The final confidence interval is:

$$0.6\pm0.0784 = (0.5216,\ 0.6784)$$

This means:

With 95% confidence, the true proportion of university students taller than 170 cm is between 52.16% and 67.84%.

mistake spotting

Learners sometimes forget that a confidence interval is not just the point estimate, but also requires adding and subtracting the margin of error to create the interval range.

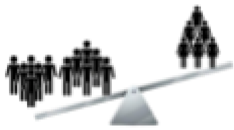
Verify Your Learning!

1. A researcher wants to estimate the proportion of customers who prefer a new product. She collects a random sample and calculates a 95% confidence interval for the proportion. What does the confidence interval represent, and how should the researcher interpret the results?
2. A survey of 200 employees found that 120 prefer remote work. Using a 95% confidence level (with a t-multiplier of 1.96), calculate the confidence interval for the proportion of employees who prefer remote work.

2. Confidence Intervals for Difference in Population Proportions

Confidence intervals can compare proportions between two groups.

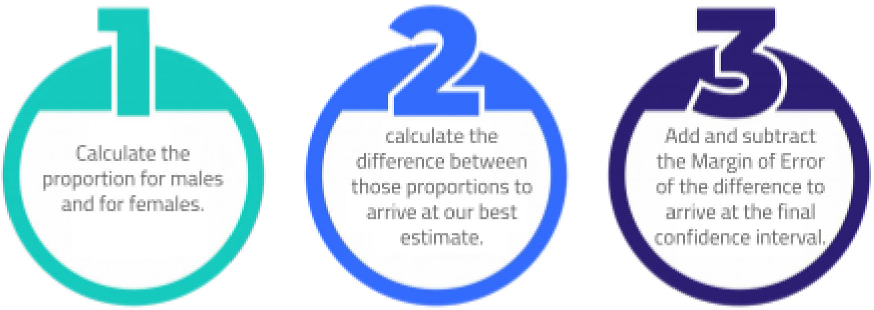
For instance, Lina wants to know if males are more likely than females to be taller than 170 cm. In other words, she wants to know the difference in proportions between the two groups.



The formula of a confidence interval is the same, but the standard error in this case is the square root of the sum of squares of each proportion's standard error.

$$SE = \sqrt{(SE_1)^2 + (SE_2)^2}$$

The complete procedure works as follows:



The following formula does exactly that:

$$\left(\hat{p}_1 - \hat{p}_2\right) \pm t^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Where:

- $\hat{p}_1, \hat{p}_2$: are the proportions for males and females respectively
- n_1, n_1: the sample size for males and females respectively

Example: Comparing proportions for tall males and females

From Lina's sample:

- Proportion of tall females $(\hat{p}_f) = \frac{40}{85} = 0.47$
- Proportion of tall males $(\hat{p}_m) = \frac{50}{65} = 0.77$

The confidence interval for the difference is:


"We all know, who is taller"

$$(0.47 - 0.77) \pm 1.96 \sqrt{\frac{0.47 \times 0.53}{85} + \frac{0.77 \times 0.23}{65}} = -0.3 \pm 0.147$$

$$= (-0.447, \ -0.153)$$

Interpretation: The proportion of tall males is 15.3% to 44.7% higher than that of tall females.

Brain power

As a data scientist, how can the choice of sample size and confidence level impact business decisions, such as customer behavior analysis or product performance evaluation? What trade-offs should be considered when balancing precision and certainty in data-driven strategies?

Verify Your Learning!

1. Why are confidence intervals important in comparing proportions, and how do they help in understanding differences between groups? What does it indicate if the confidence interval for the difference includes zero?
2. With a new sample of 60 out of 100 males and 50 out of 120 females taller than 170 cm, calculate the 95% confidence interval for the difference in proportions. Interpret the result.

3. Confidence Interval for Population Mean

Confidence intervals for the population mean estimate the true average of a quantitative variable (e.g., height) using sample data.

Standard Error of the Mean:

$$SE = \frac{s}{\sqrt{n}}$$

The full formula for the confidence interval:

$$\bar{x} \pm t^{*} \frac{s}{\sqrt{n}}$$

Where:

- $\bar{x}$: mean of the sample (best estimate)
- t^{*} : t-multiplier
- s: standard deviation of the sample
- n: sample size

Example: Average height of all students

Lina's sample data of 150 students shows an average height of 165 cm with 10 cm standard deviation.

The margin of error for a 95% confidence interval is:

$$t^{*}\frac{s}{\sqrt{n}} = 1.96\times\frac{10}{\sqrt{150}} = 1.6$$

Therefore, the confidence interval is:

$$165\pm1.6 = (163.4,\ 166.6)$$

Interpretation: With 95% confidence, the true average height of all university students is between 163.4 cm and 166.6 cm.

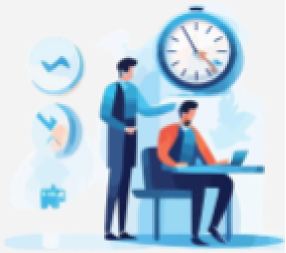
"162 cm?, stats says you're an outlier in your own university."

Confidence Intervals in Machine Learning

Fraud Detection: In fraud detection, CIs can help determine the uncertainty around the probability of a transaction being fraudulent, improving risk management strategies.

Verify Your Learning!

1. Explain the importance of using a confidence interval instead of just a sample mean when estimating a population mean. How does the confidence level (e.g., 95%) affect the interpretation of the confidence interval, and what does it imply about the uncertainty in our estimate?
2. A researcher collects a random sample of 200 employees from a company and finds that their average weekly working hours are 42 hours, with a sample standard deviation of 6 hours. Construct a 99% confidence interval for the true mean weekly working hours of all employees in the company. Clearly show your calculations and interpret the result in context.



4. Confidence Interval for the Difference in Means

Confidence intervals for the difference in means help compare averages between two groups, such as male and female students. More specifically, they help calculate a range for the difference in population means.

The Standard Error for the Difference in Means:

$$SE = \sqrt{(SE_1)^2 + (SE_2)^2}$$

Here, SE_1 and SE_2 are the standard errors of the means of each sample.

This makes the complete formula for the confidence interval:

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Where:

- $\bar{x}_1, \bar{x}_2$: means of each sample
- s_1, s_2: standard deviation of each sample
- n_1, n_2: samples sizes

Example: Comparing average heights of males and females

From Lina's sample:

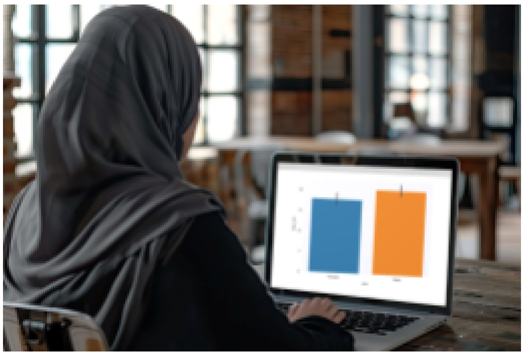
- Male height: $\bar{x}_m = 172,\ s_m = 12,\ n_m = 65$
- Female height: $\bar{x}_f = 160,\ s_f = 8,\ n_f = 85$

The confidence interval for the difference is:

$$(172 - 160) \pm 1.96\sqrt{\frac{12^2}{65} + \frac{8^2}{85}} = 12 \pm 3.4$$

$$= (8.6,\ 15.4)$$

Interpretation: With 95% confidence, the males are 8.6 cm to 15.4 cm taller than females on average.



Math in action

Confidence intervals are used in data science for two main applications:

- **Predictive Modeling**: They help assess the reliability and precision of predictions, such as churn rates in machine learning, by quantifying the uncertainty around model outputs.

- **A/B Testing**: Confidence intervals are used to determine if differences between groups (e.g., website designs) are statistically significant, helping businesses make informed decisions.

Confidence Levels

In our examples we were using the **95%** confidence and the appropriate t-multiplier for that level was 1.96. This is indeed the most commonly used confidence level. However, here is a table of other common confidence levels and their respective t-multipliers. Please refer to the t-distribution table at the beginning of the chapter for additional details.

Confidence Level	t-multiplier
90%	1.645
95%	1.96
99%	2.576

Verify Your Learning!

1. How does the standard error contribute to the calculation of the confidence interval for the difference in means, and why is it important to consider both sample sizes and standard deviations?
2. Given the following data:
   Group 1: mean = 85, standard deviation = 14, sample size = 30
   Group 2: mean = 78, standard deviation = 10, sample size = 40
   Calculate the 95% confidence interval for the difference in means and explain the result.

Explore the world of statistics with Python through Colab notebook—to get started, access it here: [CI.ipynb - Colab]

How This Concept Shapes My Field

A Chilling What-If

Can you imagine what the world would look like without confidence intervals?

Without confidence intervals, we'd be navigating the unknown with only guesses, lacking the certainty to make informed decisions. It would be like sailing through fog, unsure of where we're headed. Confidence intervals bring clarity, helping us make sense of uncertainty and empowering us to move forward with purpose and trust.

Here's a visual to reflect that journey:

The image reflects how confidence intervals help us navigate through uncertainty, offering clarity as we move forward.

Summary

Concept	Key Formula / Notes
Confidence Interval Formula	$Best\ Estimate\ \pm\ Margin\ of\ Error$
Margin of Error	$t^{*}\ \times\ Standard\ Error$
CI for Proportion	CI $= Population\ Proportion\ or\ Mean\ \pm\ (t^{*}\ \times\ Standard\ Error)$
Standard Error of a Proportion	$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
CI for Difference in Proportions	$CI = \left(\hat{p}_1 - \hat{p}_2\right) \pm t^{*}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
Standard Error of the Mean	$SE = \frac{s}{\sqrt{n}}$
CI for Mean	$CI = \bar{x} \pm t^{*}\frac{s}{\sqrt{n}}$
SE for the Difference in Means	$SE = \sqrt{(SE_1)^2 + (SE_2)^2}$
CI for Difference in Means	$CI = (\bar{x}_1 - \bar{x}_2) \pm t^{*}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

Glossary

1. **Confidence Interval (CI)**: is a range of values, derived from a sample, used to estimate a population parameter. This interval is constructed such that if the sampling were repeated multiple times, a certain percentage of the intervals would contain the true population parameter. The width of the interval reflects the precision of the estimate.
2. **The t-multiplier (t-score)**: is used in statistics to calculate confidence intervals and perform hypothesis tests when the sample size is small (typically less than 30) and the population standard deviation is unknown. It is derived from the t-distribution, which accounts for the added uncertainty in estimates when dealing with small samples.
3. **Population Proportion (p)**: is the proportion of the population that has a particular characteristic of interest, It is used in statistics to estimate the fraction of a population that exhibits a certain behavior or trait. When estimating this from a sample, the sample proportion is used as the best estimate of p.
4. **Best Estimate**: is the most likely value for a population parameter, typically derived from the sample data. The sample mean, for example, is often used as the best estimate for the population mean. The best estimate is considered the most accurate point estimate, assuming no bias in the data.
5. **Margin of Error (MoE)**: is the amount of random sampling error in a statistical estimate, typically expressed alongside a confidence interval. It indicates the range within which the true population parameter is likely to lie with a given confidence level.
6. **Standard Error (SE)**: measures the variability of a sample statistic (e.g., the sample mean) from the true population parameter. It is the standard deviation of the sampling distribution of a statistic and provides an estimate of the precision of the sample estimate.
7. **Confidence Level (CL)**: is the probability that a confidence interval will contain the true population parameter if the sampling process is repeated numerous times. Common confidence levels include 90%, 95%, and 99%. A higher confidence level corresponds to a wider confidence interval.

8. **Degrees of Freedom (df)**: refers to the number of independent pieces of information available for the estimation of a parameter. In the context of the sample variance or standard deviation, degrees of freedom reflect the number of independent data points that are free to vary when estimating a population parameter.

References

Visual Activity

The equations we learned are already programmed into many common statistical tools and libraries in Python or other analysis tools. However, it is important to understand the relationship between the confidence interval and the parameters.

For that purpose, scan this QR code:

SCAN ME

Or through this link: Confidence intervals of the mean

How to play:

In the interactive visualization, you are creating samples of fish and measuring their lengths. The goal is to estimate the average length of the entire population from the sample using a confidence interval. At 95% confidence and with repeated sampling, you can see that about 95% percent of the calculated confidence intervals (one per sample) contain the actual population mean.

You can play around with the mean, standard deviation, sample size, and confidence level in order to see the effect of those parameters on the length of the confidence interval of each sample and the success rate (the rate of confidence intervals of samples that contain the actual population mean).

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